```java
public int findMax(int[] arr) {
    int low = 0, high = arr.length - 1;

    while (low < high) {
        // If the current sub-array is already sorted,
            the last element is max
        if (arr[low] <= arr[high]) return arr[high];

        int mid = low + (high - low) / 2;

        // If the left half is sorted, the pivot is in
            the right half
        if (arr[mid] >= arr[low]) {
            low = mid + 1;
        } else {
            // If the left half is not sorted, the
                pivot is in the left half
            high = mid;
        }
    }
    return arr[low];
}
```

```
Element found at index: 5

=== Code Execution Successful ===
```

Java Implementation (Rotated Array)

Java

```java
public int findMax(int[] arr) {
    int low = 0, high = arr.length - 1;

    while (low < high) {
        // If the current sub-array is already sorted, t
        if (arr[low] <= arr[high]) return arr[high];

        int mid = low + (high - low) / 2;

        // If the left half is sorted, the pivot is in t
        if (arr[mid] >= arr[low]) {
            low = mid + 1;
        } else {
            // If the left half is not sorted, the pivot
            high = mid;
        }
    }
    return arr[low];
```

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